

Software Requirements Specification (SRS)

Project: GenAI Furniture Recommender

1. Introduction

The **GenAI Furniture Recommender** is a Retrieval-Augmented Generation (RAG) based application designed to assist users in discovering furniture items through a conversational interface. Unlike traditional rule-based search engines, this system utilizes large language models (LLMs) and vector-based semantic search to understand intent, aesthetic preferences, and style constraints, providing personalized and contextually relevant recommendations.

2. Objectives

- **Semantic Understanding:** Enable users to search for furniture using natural language and descriptive style terms.
- **Contextual Recommendation:** Provide explanations for why specific items are recommended based on user preferences.
- **Data Privacy:** Ensure all processing (LLM inference and vector search) occurs locally to protect user data.
- **Localization:** Support local currency (INR) formatting for a localized user experience.

3. Functional Requirements

- **FR1: Conversational Chat Interface:** The system shall provide a chat-based UI for user interaction.
- **FR2: Semantic Retrieval:** The system shall retrieve the top-N relevant products from a catalog based on vector similarity.
- **FR3: Preference Handling:** The system shall allow users to input secondary constraints (e.g., budget, material) that influence the final recommendation.
- **FR4: Natural Language Generation:** The system shall generate human-like, conversational responses explaining the merits of the retrieved products.
- **FR5: Currency Conversion:** The system shall dynamically convert and display all prices in Indian Rupees (₹) using a fixed conversion rate.
- **FR6: Chat History Management:** The system shall maintain the state of the current conversation to support follow-up queries.

4. Non-functional Requirements

- **NFR1: Privacy:** The system must run locally using Ollama, ensuring no sensitive data is transmitted to external servers.

- **NFR2: Latency:** The retrieval process should be sub-second, and the generation process should begin within 2 seconds of the query.
- **NFR3: Accuracy:** The recommendations must be strictly grounded in the provided product catalog to avoid hallucinations.
- **NFR4: Usability:** The interface should be modern, responsive, and accessible to non-technical users.

5. System Architecture

The system follows a standard **RAG Architecture**: 1. **Data Ingestion Layer:** Converts product metadata into embeddings using a Sentence-Transformer model. 2. **Indexing Layer:** Stores embeddings in a FAISS vector database for efficient similarity search. 3. **Retrieval Layer:** Performs a semantic search against the FAISS index based on the user's query. 4. **Augmentation Layer:** Combines retrieved product context with user queries and specific prompt templates. 5. **Generation Layer:** Passes the augmented prompt to the local Mistral LLM via Ollama to generate the final response.

6. Tech Stack

- **Language:** Python 3.10+
- **LLM Provider:** Ollama (Mistral model)
- **Framework:** LangChain (Prompting, Output Parsing)
- **Vector Database:** FAISS
- **Embeddings:** HuggingFace all-MiniLM-L6-v2
- **UI:** Streamlit

7. User Flow

1. **Input:** User enters a furniture-related query in the chat input.
2. **Preference (Optional):** User provides additional constraints in the sidebar.
3. **Search:** System executes a semantic search to find the 3 most relevant products.
4. **Local Inference:** System constructs a prompt and invokes the local Mistral model.
5. **Output:** Chatbot displays the formatted recommendation in the chat window.

8. Future Scope

- **Multimodal Retrieval:** Support for image-to-image search (finding furniture from photos).
- **Personalization Engine:** Long-term memory to learn user style preferences over time.
- **Integration:** APIs for real-time stock availability and e-commerce checkout.

- **Cross-Platform:** Mobile application development using Flutter or React Native.